

ELEC 3115

Lecture 9: Week 5, Monday

200 responses submitted

What is the most important idea you are taking away from today's lecture? (Responses should reflect your own understanding.)

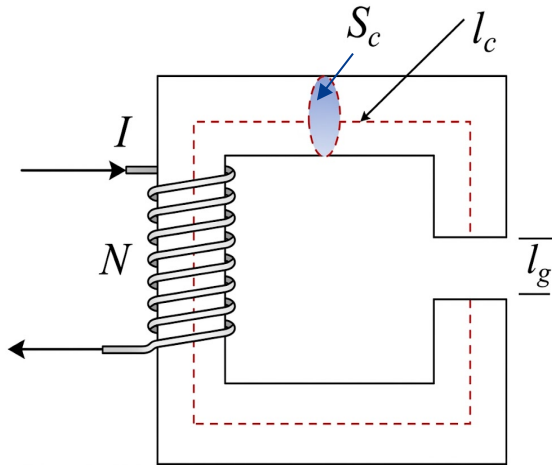
circuit and reluctance electrical circuit magnetic saturation
circuit representations electric circuits flux and current
circuit similar Circuit Model magnetic devices magnetic core
circuit and analyse **inductance** magnetic flux
flux and reluctance Magnetic Induction
air gaps **magnetic circuit**
magnetic fields
devices using a magnetic Magnetic coil

200 responses submitted

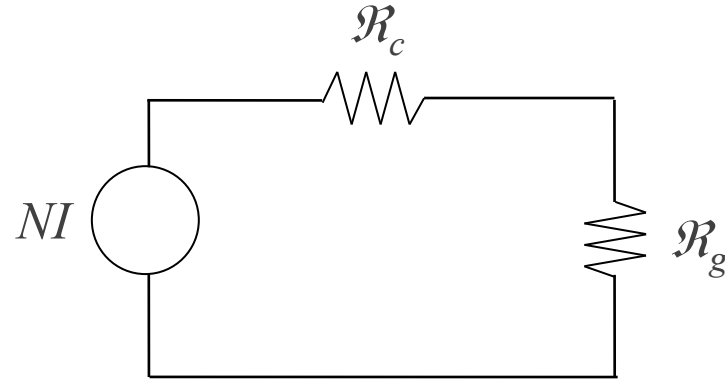
What is still unclear or confusing to you right now? (Responses should reflect your own understanding.)

Inductance calculation
Graphical method Flux Linkage graphical load
load one method Load line Air gap inductors non uniform
Inductance magnetic field
B-H graphical solution
line method B magnetic circuits
Circuit method
circuit model Magnetic devices rectangular toroid

Magnetic circuit: Reluctance (Revisit)



$$\phi = \int_s \vec{B} \cdot d\vec{s} = B \cdot S$$

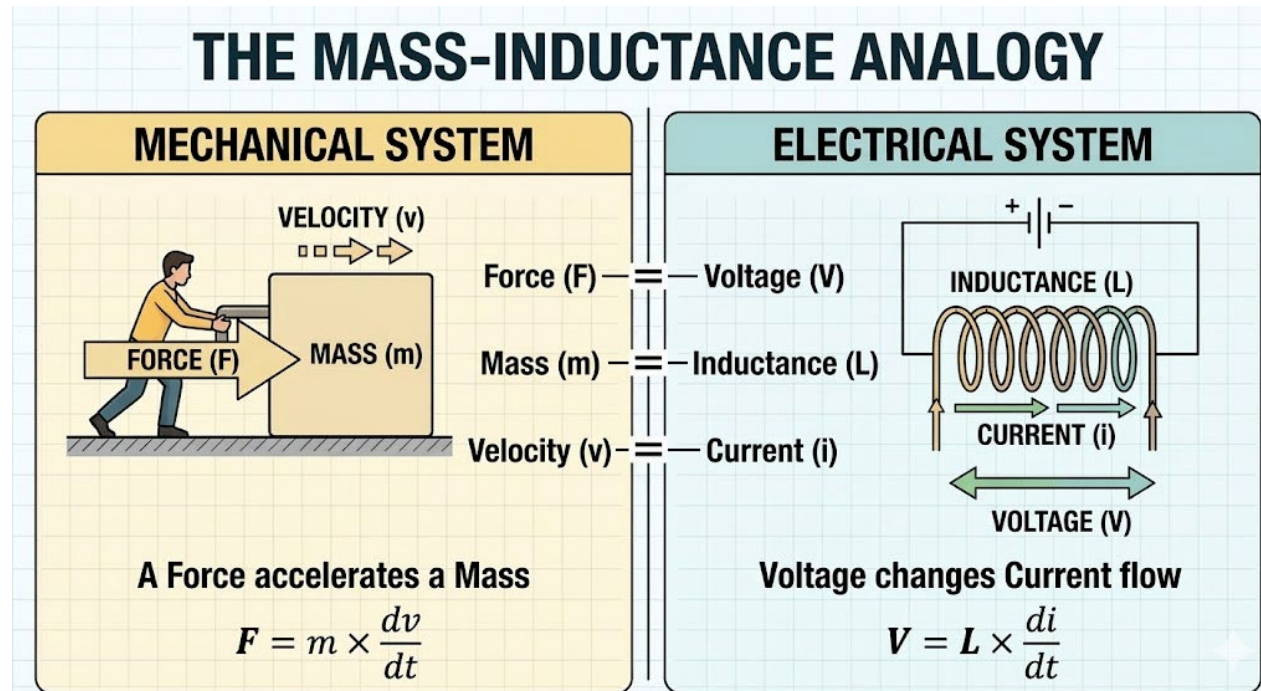
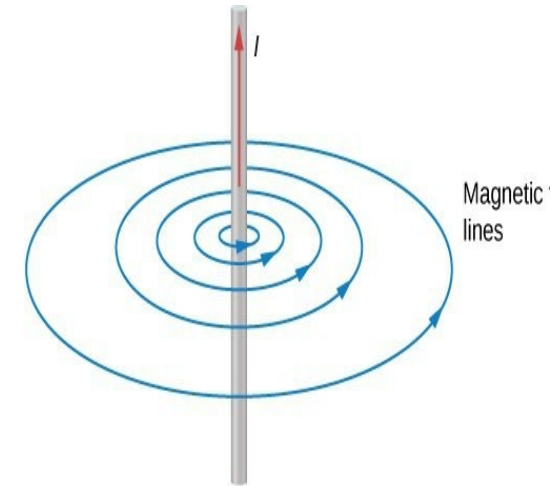


$$\mathcal{R}_c = \frac{l_c}{\mu_c S_c}, \quad \mathcal{R}_g = \frac{l_g}{\mu_o S_g}$$

Magnetic reluctance measures how difficult it is for magnetic flux to flow through a material.

Intuitive Meaning of Inductance: Revisit

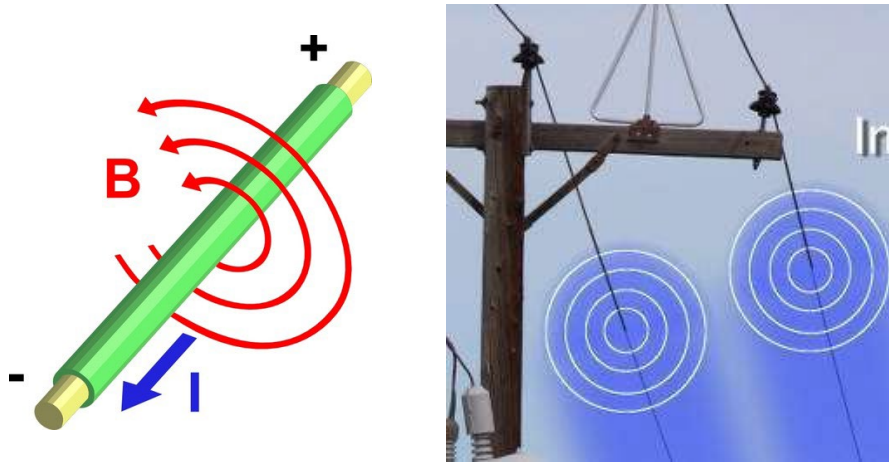
- Inductance tells us how much magnetic field a current produces and how strongly that field reacts to changes in current.
- It is also the ability of a structure to store energy in its magnetic field.



$$L = \frac{N\Phi}{I} = \frac{\lambda}{I} \text{ [Henry]}$$

👉 Keep these ideas in mind — today we will extend them.

Current-carrying conductors



- Current creates magnetic fields, and magnetic fields create inductive effects that oppose changes in current.
- Magnetic fields of near by structure can couple to each other.

Does current carrying cables and wires have inductance? How to calculate them?

Why does inductance oppose changes in current?

Current in conductor → magnetic field → flux linkage → inductance → stored energy → force

How to calculate the energy and force?

Learning goals

After today's lecture, you should be able to



analyse *inductance formation* and calculate the inductance of a current-carrying conductor/cable.



calculate mutual inductances in multi-coil/conductor systems.



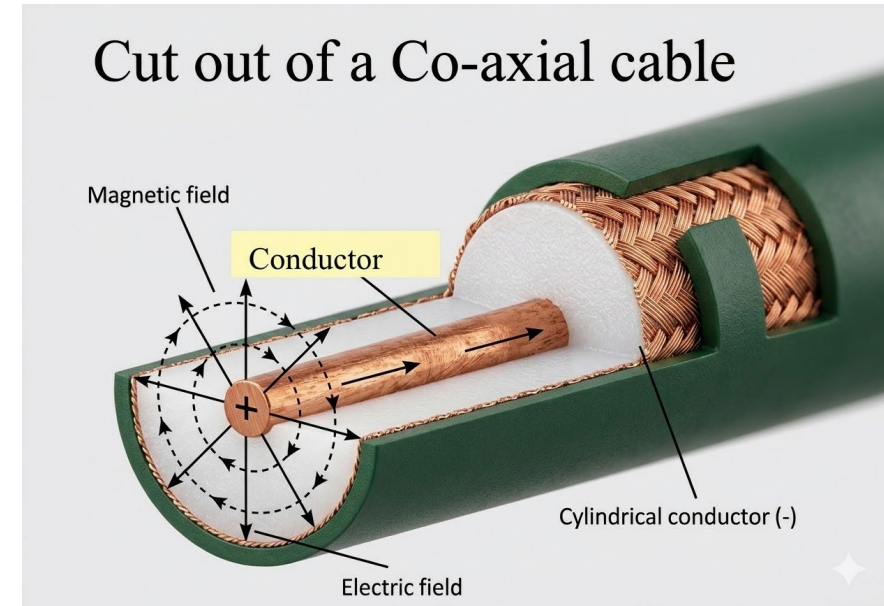
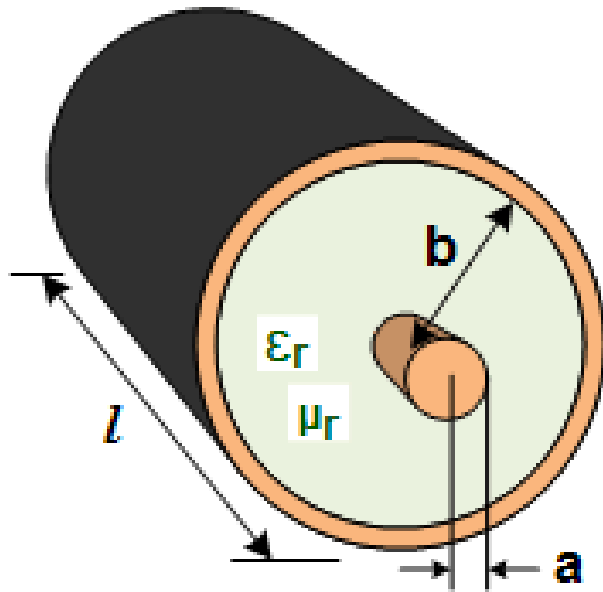
determine stored magnetic energy from $\mathbf{B}$ and $\mathbf{H}$, and calculate the related magnetic force from the energy.

Today's story: From Current → Magnetic Energy

- 1 Current in conductors creates **magnetic fields**
- 2 Magnetic fields produce **flux linkage**
- 3 Flux linkage defines **inductance**
- 4 Inductance stores **magnetic energy**
- 5 Energy gradients create **magnetic forces**

Inductance of a coaxial cable

A magnetic field is induced around the conductor of a current-carrying coaxial cable. Inductance can form in a coaxial cable.



$$B_{\text{int}} = \frac{\mu_0 r}{2\pi a^2} I, \quad 0 < r \leq a$$



Internal inductance L_{in}

$$B_{\text{ext}} = \frac{\mu_0 I}{(2\pi r)}, \quad b \geq r > a$$



External inductance L_{ext}

Internal Inductance L_{int}

Steps to calculate inductance:

$$I \rightarrow \vec{B} \rightarrow \phi = \int_s \vec{B} \cdot d\vec{s} \rightarrow \lambda = N\phi \rightarrow L = \frac{\lambda}{I}$$

$$B_{int} = \frac{\mu_0 r}{2\pi a^2} I, \quad 0 < r \leq a$$

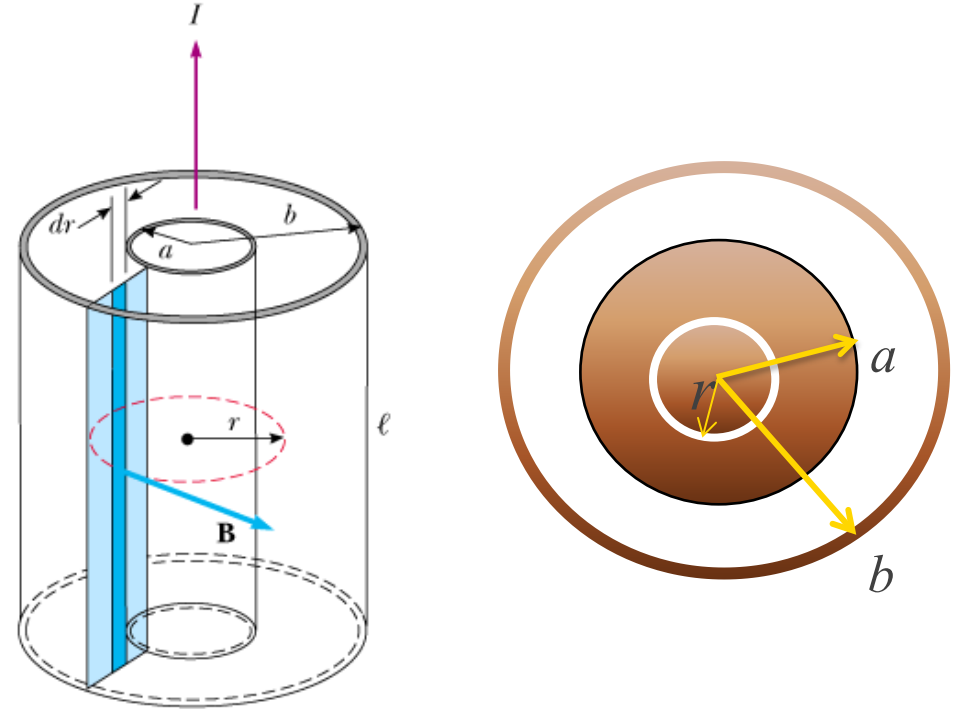
For flux calculation with non-uniform B requires defining ds on which B is normal.

$ds = l dr$ is at an arbitrary radius r inside

For the per-meter calculation, $l = 1$ m.

Flux per meter at ds :

$$d\phi_l = B_{int} ds = \left(\frac{\mu_0 r I}{2\pi a^2} dr \right) \quad (\text{wb/m})$$



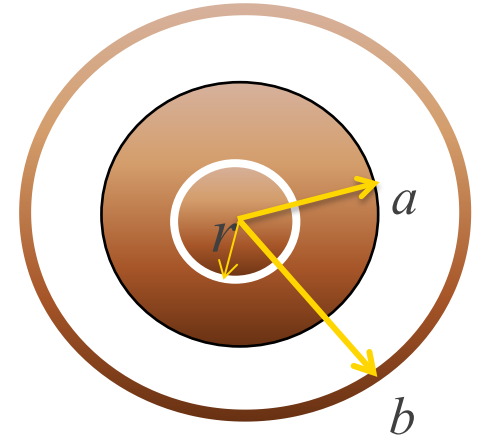
In the top view, the element ds is an annular ring of thickness dr at an arbitrary radius r . The flux linkage $d\lambda_l$ is seen by this ring.

Flux linkage per meter for this flux:

$$d\lambda_l = \Delta N (d\phi_l) = \Delta N \left(\frac{\mu_0 r I}{2\pi a^2} dr \right) \quad (\text{wb/m})$$

ΔN : Turns associated with the current in the area πr^2

$N (=1)$: Turns associated with the current in area πa^2



$$\therefore \Delta N = r^2 / a^2 \quad \text{and} \quad d\lambda_l = \frac{r^2}{a^2} \left(\frac{\mu_0 r I}{2\pi a^2} dr \right) \quad (\text{wb/m})$$

Total flux linkage per unit length:
$$\lambda_l = \int_0^a d\lambda_l = \int_0^a \left(\frac{r^2}{a^2} \right) \left(\frac{\mu_0 r I}{2\pi a^2} dr \right) = \frac{\mu_0 I}{2\pi a^4} \int_0^a r^3 dr = \frac{\mu_0 I}{8\pi}$$

Internal inductance:
$$L_{int} = \frac{\lambda_l}{I} = \frac{\mu_0}{8\pi} = \frac{4\pi \times 10^{-7}}{8\pi} = 0.5 \times 10^{-7} \quad (\text{H/m})$$

Internal inductance is a constant: 50 nH/m !!

External inductance L_{ext}

$$B_{\text{ext}} = \frac{\mu_0 I}{(2\pi r)}, \quad b \geq r > a$$

To find flux ϕ per unit length, again, we define an elementary area $ds = dr \cdot l$ at an arbitrary radial distance r , where $l = 1\text{m}$.

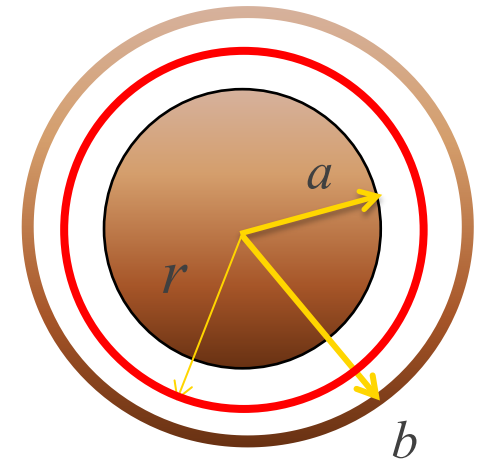
$$d\Phi_l = B_{\text{ext}} ds = \left(\frac{\mu_0 I}{2\pi r} dr \right) \quad (\text{wb/m})$$

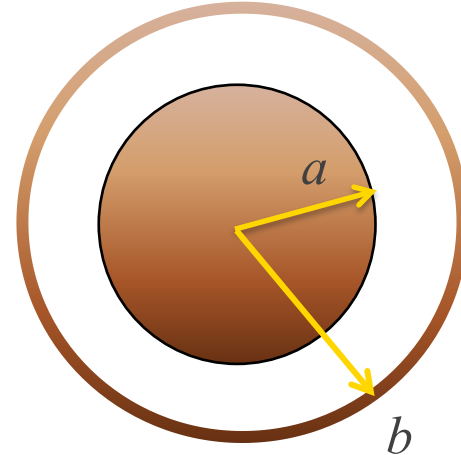
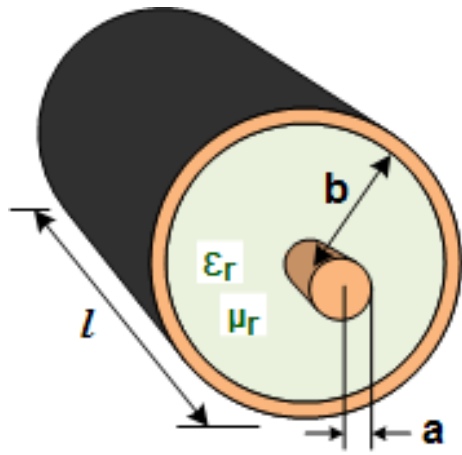
$$\text{Flux linkage for turn } N=1, \quad d\lambda_l = \left(\frac{\mu_0 I}{2\pi r} dr \right) \quad (\text{wb/m})$$

$$\text{Total flux linkage } \therefore \lambda_{\text{ext}} = \int_a^b \frac{\mu_0 I}{2\pi r} dr = \frac{\mu_0 I}{2\pi} \ln\left(\frac{b}{a}\right)$$

$$\text{Inductance: } L_{\text{ext}} = \frac{\lambda_{\text{ext}}}{I} = \frac{\mu_0}{2\pi} \ln\left(\frac{b}{a}\right) \quad (\text{H/m})$$

External inductance is a function of radii a , b and permeability $\mu = \mu_r \mu_0$





Total Inductance:
$$L = L_{int} + L_{ext} = \frac{\mu_0}{8\pi} + \frac{\mu_0}{2\pi} \ln \frac{b}{a} \quad (\text{H/m})$$

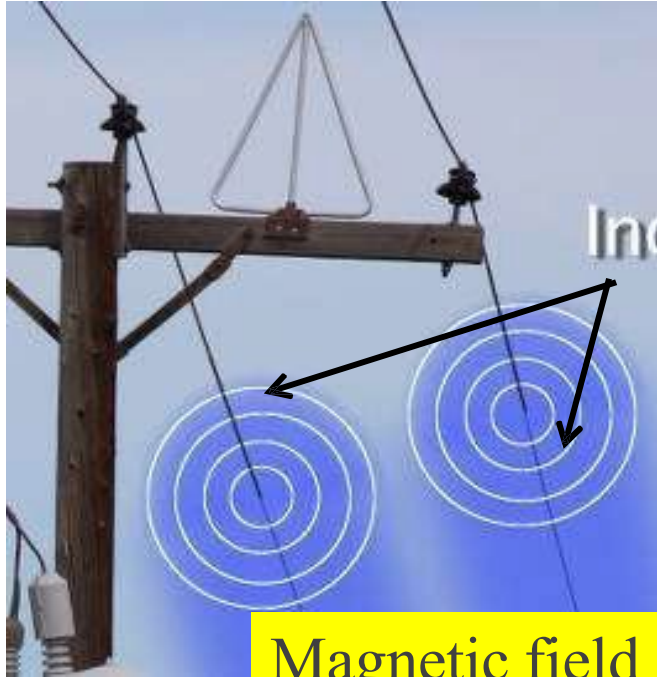
! **Exam focus**

Internal inductance, which is due to the internal field B_{in}

External inductance is due to the external field B_{out} between the radii a and b

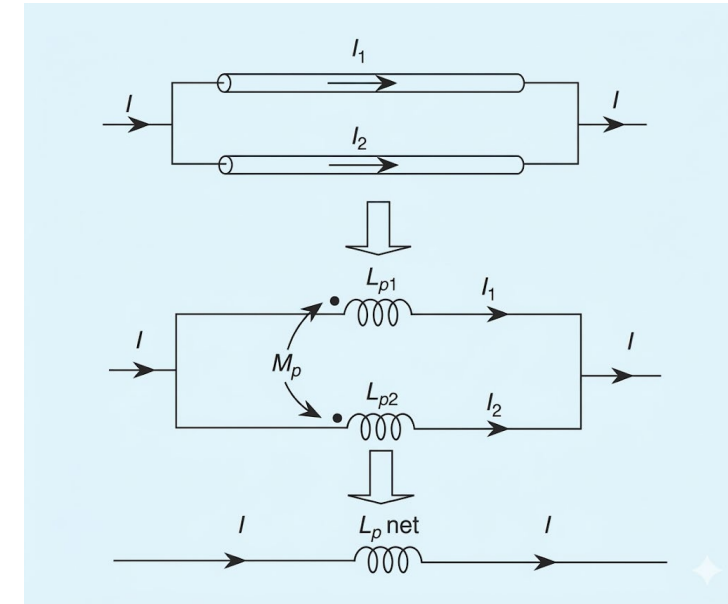
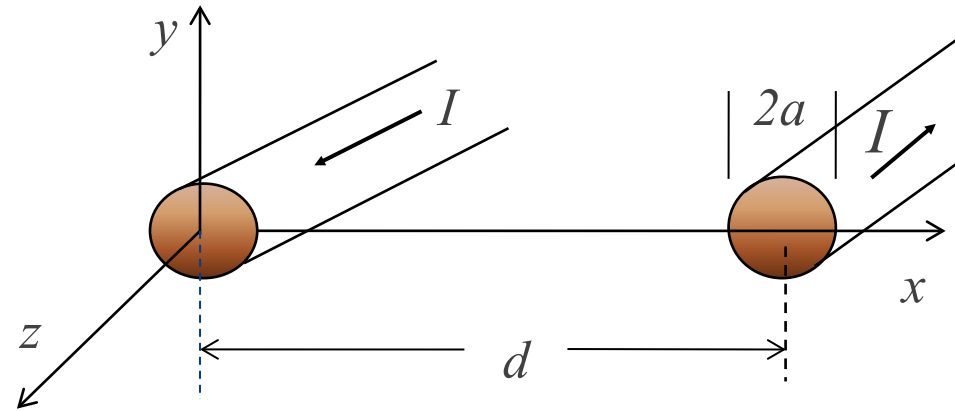
*In high frequency applications, current in a good conductor tends to be in the surface (skin) of the conductor- **skin effect**. If the current is only at the surface, the internal inductance reduces to zero. Also, in a thin conductor, the internal flux can be neglected, and then the internal inductance is neglected.*

Inductance between two parallel cylindrical conductors carrying current in opposite directions



Magnetic field

Top view:



Total internal inductance $L_i = 2 \times \frac{\mu_0}{8\pi} = \frac{\mu_0}{4\pi}$

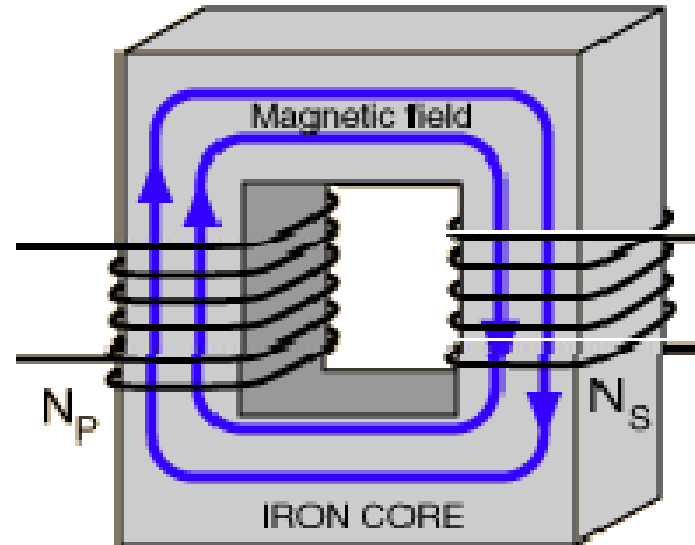
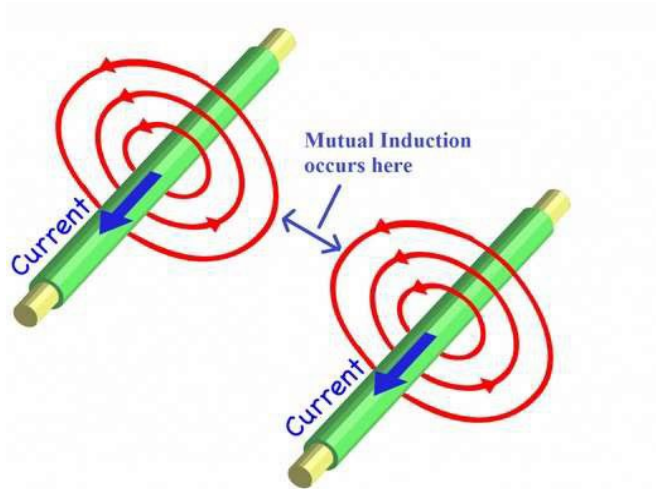
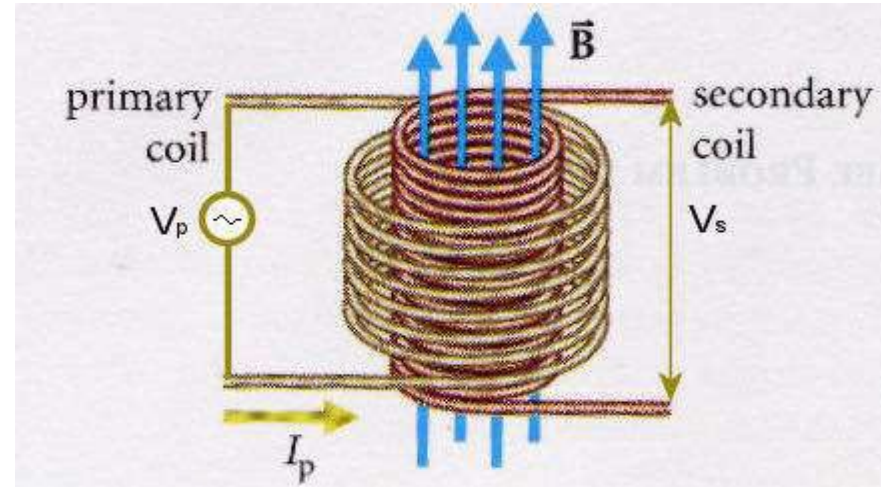
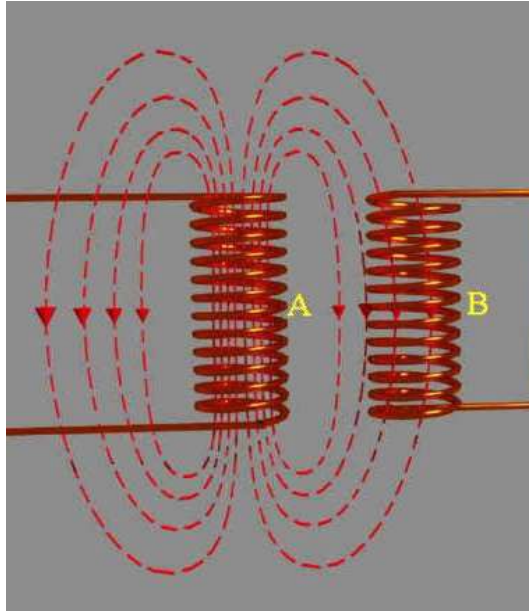
Total external inductance $L_e = 2 \times \frac{\mu_0}{2\pi} \ln \frac{d-a}{a} \approx \frac{\mu_0}{\pi} \ln \frac{d}{a}, \quad d \gg a$

Total inductance per unit length between the two-conductors $L = \left(\frac{\mu_0}{4\pi} + \frac{\mu_0}{\pi} \ln \frac{d}{a} \right) \quad (\text{H/m})$



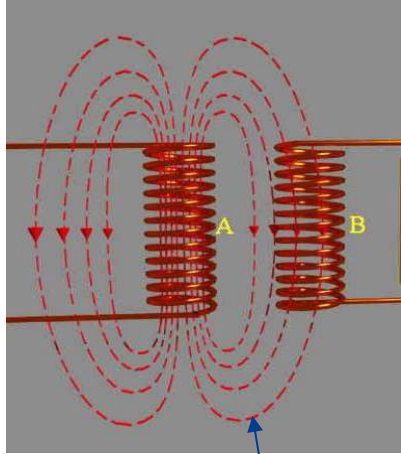
Exam focus

Mutual flux and mutual inductance



How to calculate mutual inductance?

Mutual Inductance (M)



Mutual flux Φ_{12}

Current I_1 of the coil A creates a magnetic field.
Some of the fluxes (Φ_{12}) link coil B of N_2 turns.
Shared flux (mutual flux linkage) produces mutual inductance.

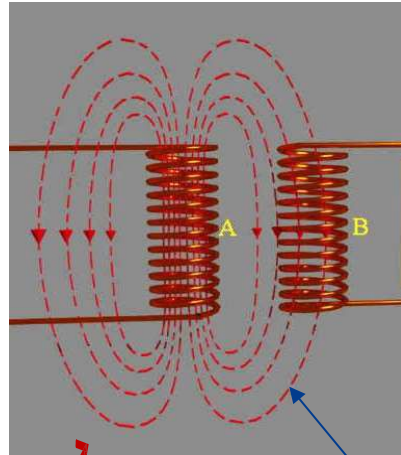
Mutual flux linkage is $\lambda_{12} = N_2 \Phi_{12}$.

The **mutual inductance** is the mutual flux linkage of one coil per unit current of the coil producing the flux.

$$M_{12} = \frac{\lambda_{12}}{I_1} = \frac{N_2 \phi_{12}}{I_1} \quad (\text{H}), \quad \phi_{12} = \int_{s_2} \vec{B}_1 \cdot d\vec{s}_2 \quad (\text{Wb})$$

A similar mutual inductance (M_{21}) occurs if the coil B also carries current.

It can be shown that $M_{21} = M_{12}$



The self-inductance of coil A is the total flux that links coil A over the current of the coil A.

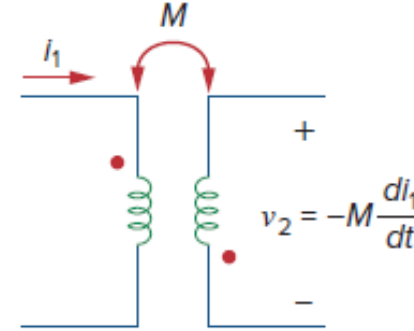
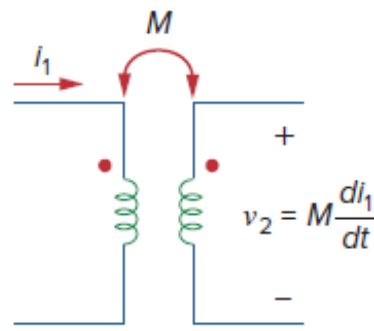
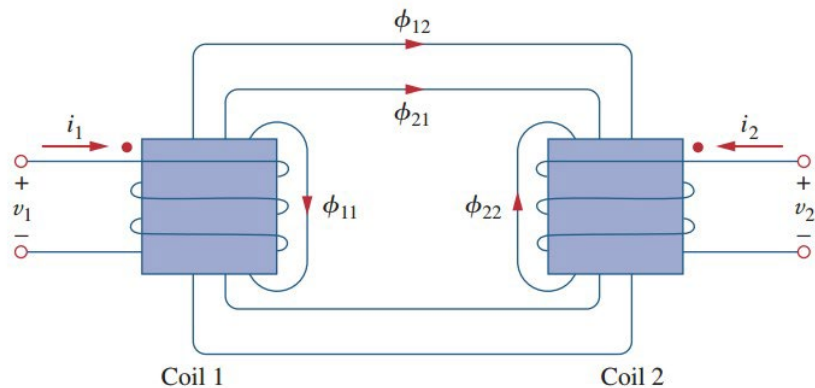
$$L_1 = \frac{\lambda_1}{I_1} = \frac{N_1 \Phi}{I_1} \quad (\text{H}), \quad \Phi = \phi_{11} + \phi_{12}$$

Leakage flux Φ_{11}

Mutual flux Φ_{12}

Flux that links only coil A is called the leakage flux ϕ_{11} .
The inductance related only to ϕ_{11} is called the leakage inductance.
 Leakage inductance comes from flux that does NOT couple.

Dot Convention for mutual inductance



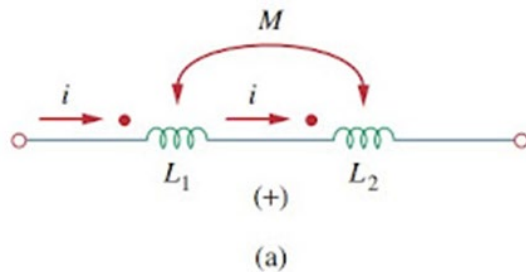
Rule 1:

Current entering dotted terminal $\rightarrow$ positive induced voltage at the other dot

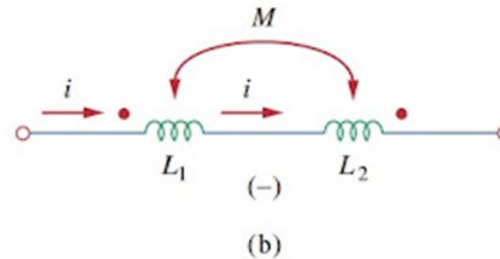
Rule 2

Equivalent inductance: Adding or opposing

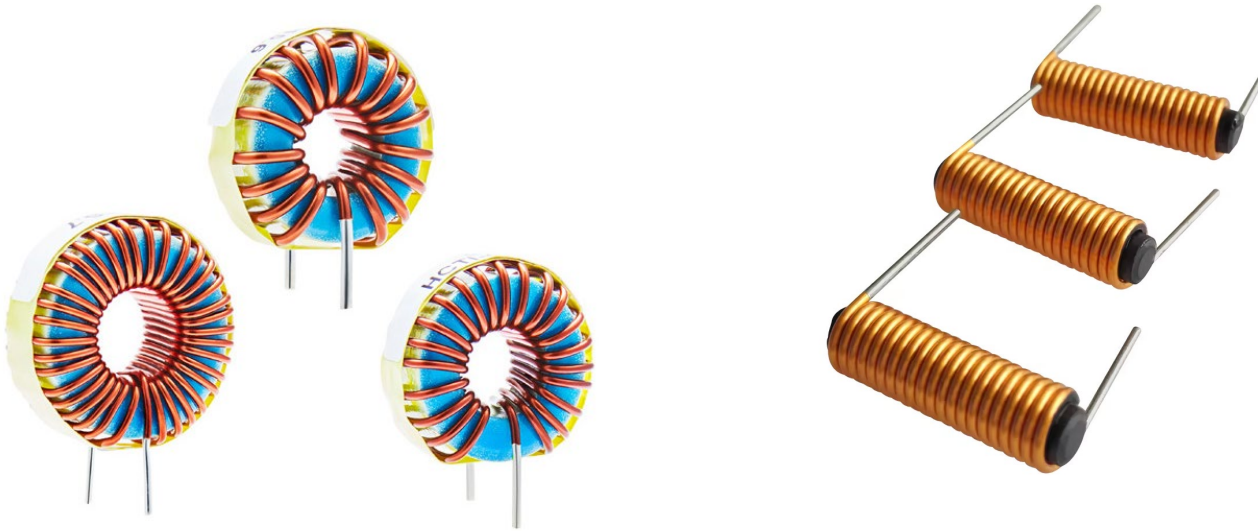
$$L_1 + L_2 + 2M$$



$$L_1 + L_2 - 2M$$



Application example 7



An application requires 150 mH, but you have 50 mH of 3 almost identical inductors. You found that the mutual inductances $M_{12} = M_{23} = 10$ mH and $M_{13} = 5$ mH.

Can you get 150 mH by connecting them in series? What are the maximum and minimum equivalent inductances possible?

How will you determine the dot convention if not indicated by the manufacturer?

Is there any way to get 150 mH from the three 50 mH inductors?

$$L_{max} = L_1 + L_2 + L_3 + 2M_{12} + 2M_{23} + 2M_{13}, \quad L_{min} = L_1 + L_2 + L_3 - 2M_{12} - 2M_{23} + 2M_{13}$$

Pause & Reflect

(Before we take a short break...)

 **Take 1 minute to jot down:**

One idea you just learned

One step you could explain to a friend

One word that describes how confident you feel right now
(scale: 😞 *unsure* → 😊 *okay* → 😄 *confident*)

💡 *This is for you — no submission.*

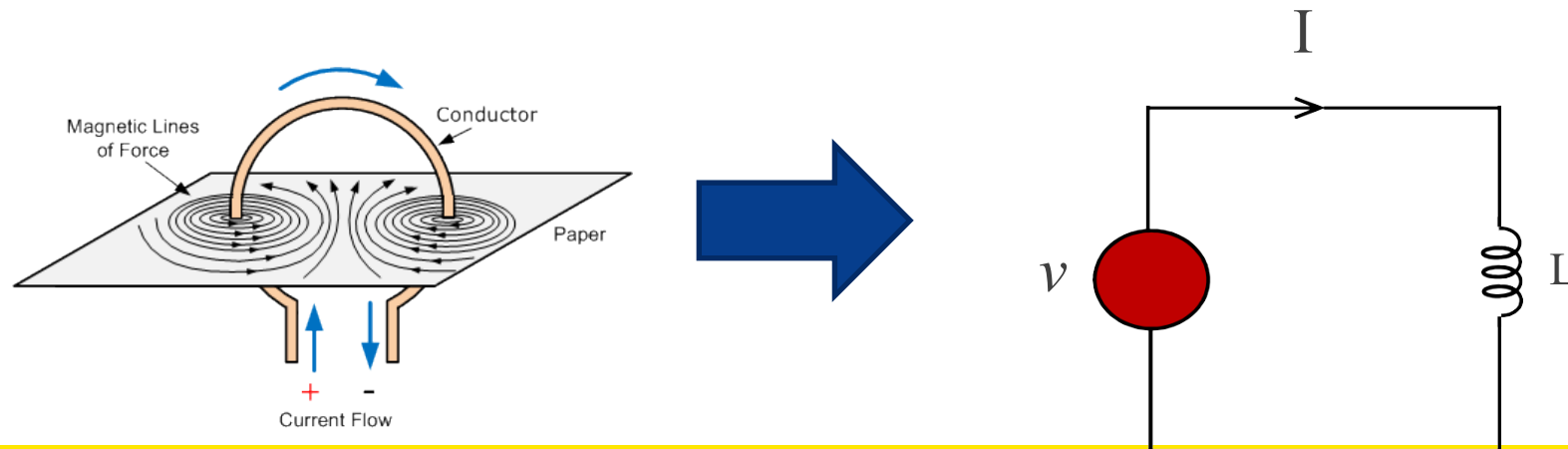


Stored energy in a magnetic field

Stored energy in an inductor: $W_m = \frac{1}{2}LI^2$ (J)

From where does this energy come?

In a loss-less ideal system, energy of the source is spent to send the current I . This spent energy is stored in the magnetic field of the system i.e., in the inductance of the system.



$$W_m = \frac{1}{2} L I^2, \text{ and } L = \frac{\lambda}{I}$$

$$\text{Substituting, } \therefore W_m = \frac{1}{2} \lambda I = \frac{1}{2} (N\phi) I$$

It is useful to express stored magnetic energy in terms of **B** and **H**.

$$\phi = \int \vec{B} \cdot d\vec{s}, \quad \oint_c \vec{H} \cdot d\vec{l} = NI \quad \longrightarrow \quad W_m = \frac{1}{2} \int_{vol} \vec{B} \cdot \vec{H} \, dv \quad (J)$$

$$\text{Also, magnetic energy density (energy per unit volume): } w_m = \frac{1}{2} \vec{B} \cdot \vec{H} \quad (\text{J/m}^3)$$

$$\therefore \vec{B} = \mu \vec{H} \quad w_m = \frac{1}{2} \vec{B} \cdot \vec{H} = \frac{B^2}{2\mu} \text{ or } w_m = \frac{1}{2} \vec{B} \cdot \vec{H} = \frac{\mu}{2} H^2$$

Magnetic Force

Like electrostatic force, magnetostatic force can also be found from stored magnetic energy by using *virtual displacement principle*.

$$\vec{F}_m = \pm \vec{\nabla} W_m \quad (\text{N}), \quad W_m = \frac{1}{2} LI^2 = \frac{1}{2} \lambda I = \frac{1}{2} \int_{vol} \vec{B} \cdot \vec{H} \, dv \quad (J)$$

$$\vec{F}_m = \vec{\nabla} W_m \Big|_{\text{constan } I}, \quad \vec{F}_m = -\vec{\nabla} W_m \Big|_{\text{constan } \lambda}$$

If a source is present i.e. current, I is constant, $\vec{F}_m$ is positive gradient of W_m .

If there is no source to replenish the spent energy of virtual work done, stored energy reduces i.e., force is the negative gradient of W_m . Absence of source also means that there is no change to the flux linkage i.e. λ constant system.

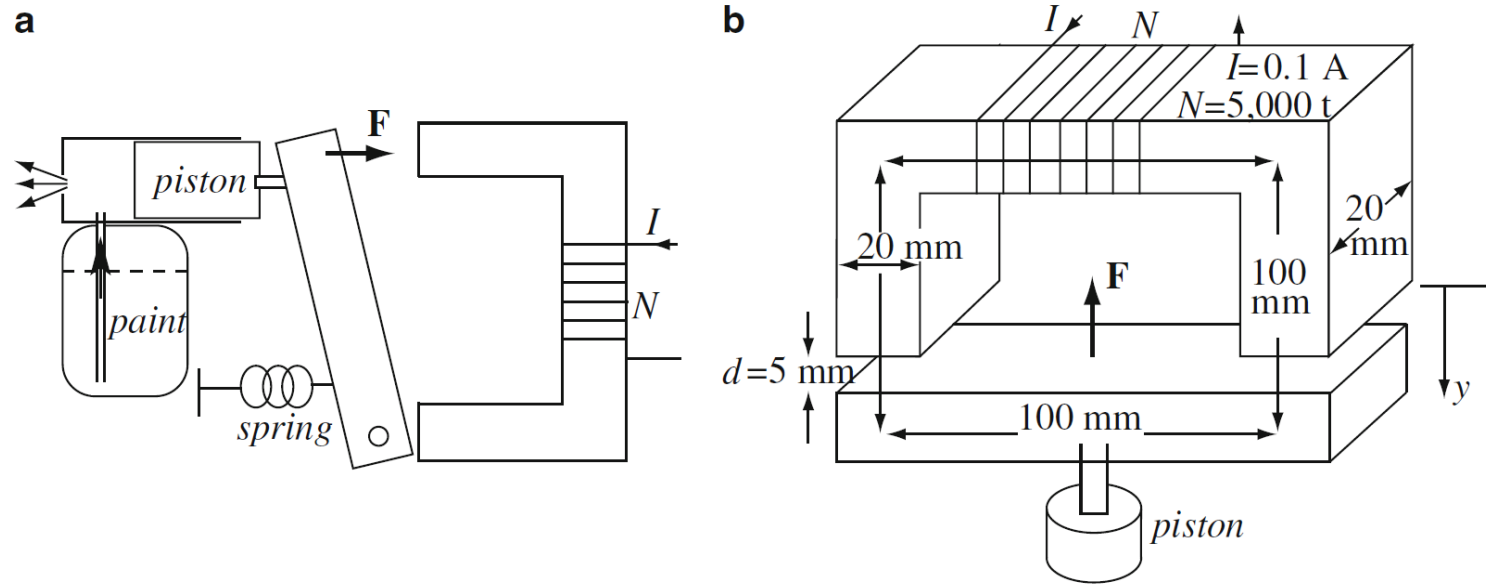
Cartesian co-ordinate: $\vec{F}_x = \pm \frac{\partial W_m}{\partial x} \vec{a}_x, \quad \vec{F}_y = \pm \frac{\partial W_m}{\partial y} \vec{a}_y, \quad \vec{F}_z = \pm \frac{\partial W_m}{\partial z} \vec{a}_z$



Example 8

An electric pump for spray paint is shown in the figure. The coil contains $N=5000$ turns and carries a current $I = 0.1\text{ A}$. The permeability of iron is $\mu=1000\ \mu_0$, and the air gap is $d=5\text{ mm}$ long. Assume no leakage flux in the gap. All other dimensions can be seen in the second figure. Calculate the force exerted by the moving piece on the piston.

[Ans: 1.146 N]



Solution

$$\vec{F}_m = \pm \vec{\nabla} W_m \quad (\text{N}), \quad W_m = \frac{B_g^2}{2\mu_0} \times \text{volume}$$

The approach to solve this problem is –

1. Find the flux density B_g at the air-gap from the magnetic circuit.
2. Find stored energy in the gap as $W_m = \frac{1}{2\mu_0} B_g^2 \times \text{volume of the gap}$
3. Find force $F_y = \pm \frac{dW_m}{dy}$

Draw the magnetic ckt:

$$\mathfrak{R}_c = \frac{l_c}{S_c \mu_c} \quad \mathfrak{R}_g = \frac{l_g}{S_g \mu_0}$$

$$\Phi = \frac{NI}{(\mathfrak{R}_{Tot})} \rightarrow B_g = \frac{\Phi}{S_g} = 0.06T$$

Total energy stored in one of the two gaps $= \frac{1}{\mu_0} B_g^2 \times Sy$

Airgap length d varies by the force in the y direction, so d becomes a variable. We need to find force exerted on the piston (i.e., only at the one airgap).

$$F_y = \frac{d(W_m)}{dy} = \frac{1}{2\mu_0} B_g^2 \times S = 0.582 \text{ N}$$

(The spring exerts exact opposite force in the other air-gap to hold the moving part.)

Alternatively, we could find inductance L in terms of total reluctance,

$$L = \frac{N^2}{\text{Reluctance}}$$

Find energy $W_m = \frac{1}{2} LI^2$ and force $F_y = \frac{dW_m}{dy}$

Try at home - check the numerical value is same with this method.

Before You Go... (1 minute reflection)

This reflection will count toward optional bonus marks.



0 response submitted

What is the most important idea you are taking away from today's lecture? (Responses should reflect your own...)



Waiting for response...

Responses will be displayed as a word cloud

Wordcloud

All responses

<

1 of 3

>

Scan the QR or use link to join



<https://forms.office.com/r/vJs4eSWdQH>

 Copy link

Key ideas

- ✓ Inductance comes from flux linkage
- ✓ Geometry determines inductance
- ✓ Mutual inductance links coils
- ✓ Magnetic fields store energy
- ✓ Energy gradients produce forces

More examples for review (recommended)

Examples 6-15, 6-18, 6-21, 6-23 of your textbook by D.K. Cheng.